

ATmega328P supports **six sleep modes**

1. **Idle** – CPU stops; peripherals (Timers, USART, etc.) keep running. This mode has the **fastest wake-up**.

Wake-up Sources: Any enabled interrupt, such as from a Timer, USART, or Pin Change, can wake the CPU.

When the SM2..0 bits are written to 000, the SLEEP instruction makes the MCU enter idle mode

2. **ADC Noise Reduction** – CPU and I/O stops; ADC runs with less noise.

Wake-up Sources: ADC interrupt, external interrupt, watchdog

When the SM2..0 bits are written to 001, the SLEEP instruction makes the MCU enter ADC noise reduction mode

3. **Power-down** – Most peripherals are turned off; This consumes the lowest power.

Wake-up Sources: External interrupt (INT0/1, Pin Change), watchdog time

When the SM2..0 bits are written to 010, the SLEEP instruction makes the MCU enter power-down mode

4. **Power-save** – Like Power-down, but Timer2 runs asynchronously for timed wake-ups.

Wake-up Sources: Timer2 overflow, external interrupt, watchdog

When the SM2..0 bits are written to 011, the SLEEP instruction makes the MCU enter power-save mode

5. **Standby** – Like Power-down, with faster wake-up using the external oscillator.

Wake-up Sources: External interrupt, watchdog

When the SM2..0 bits are 110 and an external crystal/resonator clock option is selected, the SLEEP instruction makes the MCU enter standby mode

6. **Extended Standby** – Combines Power-save and Standby for both Timer2 and fast wake-up.

Wake-up Sources: Timer2, external interrupt, watchdog

When the SM2..0 bits are 111 and an external crystal/resonator clock option is selected, the SLEEP instruction makes the MCU enter extended standby mode. This mode is identical to power-save with the exception that the oscillator is kept running

Registers

SMCR – Sleep Mode Control Register

The sleep mode control register contains control bits for power management. Bit 0 – SE: Sleep Enable The SE bit must be written to logic one to make the MCU enter the sleep mode when the SLEEP instruction is executed

Sleep enables bits

- **Bits 3..1 – SM2..0: Sleep Mode Select Bits 2, 1, and 0**

These bits select between the five available sleep modes as shown in [Table 9-2 on page 38](#).

Table 9-2. Sleep Mode Select

SM2	SM1	SM0	Sleep Mode
0	0	0	Idle
0	0	1	ADC noise reduction
0	1	0	Power-down
0	1	1	Power-save
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Standby ⁽¹⁾
1	1	1	External standby ⁽¹⁾

Note: 1. Standby mode is only recommended for use with external crystals or resonators.

- **Bit 0 – SE: Sleep Enable**

The SE bit must be written to logic one to make the MCU enter the sleep mode when the SLEEP instruction is executed. To avoid the MCU entering the sleep mode unless it is the programmer's purpose, it is recommended to write the sleep enable (SE) bit to one just before the execution of the SLEEP instruction and to clear it immediately after waking up.